Chit 9 — MySQL + MongoDB Mixed Queries

MySQL Queries

i. Names of instructors with courses taught semester-wise

```
SELECT i.name, t.semester, t.year, c.title
FROM instructor i
JOIN teaches t ON i.T_ID = t.T_ID
JOIN course c ON t.course_id = c.course_id
ORDER BY i.name, t.year, t.semester;
```

ii. Create View on student table

```
CREATE VIEW student_view AS
SELECT S_ID, name, dept_name, tot_cred
FROM student;

SELECT * FROM student_view;
```

iii. Rename column dept_name to department_name

```
ALTER TABLE student
CHANGE dept_name department_name VARCHAR(50);
```

iv. Delete students whose department is NULL

```
DELETE FROM student
WHERE department_name IS NULL;
```

MongoDB — orderinfo Collection

```
// Create and insert documents
use lab_db
db.orderinfo.insertMany([
  { cust_id: 123, cust_name: 'abc', status: 'A', price: 250 },
  { cust_id: 124, cust_name: 'xyz', status: 'B', price: 500 },
  { cust_id: 125, cust_name: 'pqr', status: 'A', price: 750 },
  { cust_id: 126, cust_name: 'mno', status: 'A', price: 150 }
])
```

i. Average price for each customer with status 'A'

```
db.orderinfo.aggregate([
  { $match: { status: 'A' } },
  { $group: { _id: '$cust_id',
              avg_price: { $avg: '$price' } } }
])
```

ii. Status of customers whose price is between 100 and 1000

```
db.orderinfo.find(
  { price: { $gte: 100, $lte: 1000 } },
  { _id: 0, cust_name: 1, status: 1, price: 1 }
)
```

iii. Display customer info without _id

```
db.orderinfo.find({}, { _id: 0 })
```

iv. Create simple index on orderinfo and fire queries

```
// Create index on cust_id
db.orderinfo.createIndex({ cust_id: 1 })

// Verify index
db.orderinfo.getIndexes()

// Fire query using index
db.orderinfo.find({ cust_id: 123 })
```